

Expert systems in medicine

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Abstract

Medical expert systems (ESs) aim to apply computer technology to emulate human decision-making and provide computerized clinical decision support to clinicians, patients, and other individuals with suitable information and knowledge at appropriate times to improve the quality and safety of health care. This chapter first presents the significance and a brief history of this field. It then describes a common architecture and major components of medical ESs and introduces different knowledge representation and reasoning techniques. This chapter will also show some examples of medical ESs, including computer-assisted diagnosis systems, medication alert systems, reminder systems, and so on. Lastly, it discusses the advantages and disadvantages of existing approaches as well as major issues and challenges related to system implementation, evaluation, maintenance, and distribution and points out some of the directions for future research and development.

Keywords: Expert systems; knowledge bases; rule-based systems; clinical decision support; decision theory; knowledge acquisition; knowledge representation; computer-assisted diagnosis; computer-assisted therapy; medication alert system; reminder systems

5.1 Introduction

In medicine and health care, many computer systems have been developed to automatically assist clinicians and patients with decision-making by providing relevant information in a timely manner. The use of artificial intelligence (AI) and expert systems (ESs) as an aid to clinicians was first recognized in the late 1950s. Medical ESs aim to apply computer technology to emulate human decision-making and to provide computerized clinical decision support (CDS) to clinicians, patients, and other individuals with pertinent information and knowledge at appropriate times to improve health and health-care delivery. Medical ESs typically consist of a knowledge base and an inference engine. The inference engine reasons upon available knowledge from the knowledge base and input data to make health-care recommendations. The user interacts with the system through a user interface, where the user can have an interactive dialog with the system or enter information if needed, and then receives recommendations and interpretations of these recommendations given by the system. The purpose of these systems is to assist users to make better

analyses or decisions based on the available data and knowledge than either the user or the system could make on their own.

It is widely recognized that ESs and other CDS systems have great potential to improve health and health care. Medical errors are common and costly and can result in death and serious injury. The Institute of Medicine report *To Err Is Human*, issued in 1999, estimated that as many as 98,000 people died in hospitals per year from medical errors,¹ while many injuries that occur during hospitalization are unpredictable and unavoidable, 20%–70% may be preventable. Computer information technology, such as ESs, is believed to provide optimal evidence-based care to make the best knowledge available when needed and help one to reduce medical errors. ESs can aid at building a safer health by providing more precise diagnosis and a more scientific determination of the treatment plan, check potential medication errors at each stage of the medication management process (e.g., ordering, dispensing, and administering), and inform clinicians of actionable interventions based on patient conditions.

In this chapter, we first present a brief history of this field and then introduce a common architecture and major components of medical ESs, as well as different knowledge representation (KR) and reasoning techniques. We will demonstrate some examples of medical ESs and discuss the advantages and disadvantages of existing approaches. We will also present major issues and challenges in this field and finally point out future directions.

5.2 A brief history

After the dawn of modern computers in 1950s, medical researchers began experimenting with the idea of using computer technology to emulate the decision-making processes of clinical domain experts. Clinical diagnostic systems were the early forms of ESs. In the late 1950s, Ledley and Lusted proposed the use of mathematical techniques to model the reasoning process of medical diagnosis.² In the 1970s and 1980s the field of medical AI and ESs was explored and established.³ Some exemplary medical ESs were developed. During this time, MYCIN was designed to identify bacteria causing infection and recommend antibiotics.⁴ However, most early diagnostic consultation systems were limited to stand-alone applications or working prototypes and were not used in routine clinical practice due to many obstacles, including integration with the hospital information systems and clinician workflow, the complexity and uncertainty of the logic rules, the high costs of maintaining the knowledge base, and the requirement of manual data entry and cumbersome dialogs.⁵

During the late 1980s and early 1990s, excitement surrounding AI fell into what was often referred to as the “AI winter.” Over the years, researchers in medical informatics, AI, and other related fields have strived to find solutions to address and overcome the limitations of ESs. These efforts included developing KR frameworks and common reference models, markup languages (such as Arden Syntax⁶ for presenting and sharing medical knowledge in an executable format), and computer-based medical coding system (such as GALEN that aimed to provide reusable terminology resources for clinical systems).⁷ To address uncertainty, which is common in medicine, fuzzy logic was integrated into languages,

such as fuzzy Arden Syntax.⁸ Evaluation studies, including randomized, controlled trials, were conducted to assess provider performance, clinical outcomes, and economic impact of the ESs.^{9–12} It was also realized that some potential benefits, such as improved process of care, quicker and easier access to patient data, and automatic error checks, often cannot be measured directly by clinical outcome measures.

In the late 1990s and early 2000s, with the advancement and increased use of the Internet and information technology in health care and the emphasis on evidence-based medicine (EBM) and clinical guidelines, more ESs and computerized CDS systems emerged. While the majority of early ESs focused on diagnosis, new areas, including health-care reminders and medication alerts, began to be further explored. With the rapid increase in the amount of new information available in health care, knowledge management became an important task. Efficient methods and tools were critical to allow medical domain experts, knowledge engineers (KEs), and software engineers to convert medical knowledge into machine executable decision support rules.¹³ Query and expression languages, such as GELLO,¹⁴ were developed for specifying logic expressions and criteria to facilitate representation and sharing of computerized clinical knowledge.

Electronic health record (EHR) technologies progressed rapidly and saw widespread adoption in the late 2000s. It has been widely recognized that the addition of expert knowledge that can provide computerized CDS in the EHR is critical to facilitating meaningful use of the EHR as both a patient care and population management tool, rather than as a platform primarily for billing and documentation. However, many health-care delivery institutions lacked resources and expertise to implement advanced expert and CDS systems to realize their potential benefits. In addition, institutions often implemented these systems in different ways due to the lack of a universally adopted standard representation of the best available clinical knowledge for knowledge sharing and distribution. In the mid-2000s a road map was proposed that recommended a series of actions to improve CDS capabilities and increase its use through the US health sector.¹⁵ A set of coordinated, collaborative projects was funded and conducted¹⁶ to develop approaches for representing structured and shareable CDS intervention artifacts¹⁷ and distributing the machine executable artifacts via web-services.¹⁸

Over the past five decades, despite notable successes of medical ESs and CDS, wide adoption and efficient use of such systems still face many challenges. The critical path ahead includes, but is not limited to, improving KR, interoperability and sharing, increasing integration with workflow and effective use, and leveraging data to enhance knowledge, reasoning processes and methods.

5.3 Methods

5.3.1 Expert system architecture

An ES is a knowledge-based system that consists of a knowledge base from a specific, usually narrow, domain, and an inference engine to reason about that specific domain. An ES interacts with users through a user interface. It simulates the decision-making processes human experts (HEs) follow to tackle specific problems.

ESs find their way into most areas of knowledge and are used in a wide variety of applications, including medical diagnosis, helping physicians identify potential diagnoses based on signs and symptoms, and patient data^{19,20}; planning²¹ and scheduling in airline and cargo applications²²; configuration applications with constraints²³ to maximize usability of resources²⁴; design and manufacturing²⁵; financial decision-making in the stock market and fraud detection²⁶; and process management to maximize the usability of resources while managing costs.²⁷ Despite the apparent disparate nature of applications, what all these areas have in common is that knowledge is abstracted from the real world and represented in a concrete, unambiguous manner in the form of facts and rules with a tractable number of possible solutions.

Building an ES starts with a HE, a KE, and a user (U). The first step is to define the problem at hand to address the needs of the user. This requires the concerted effort of HE, KE, and U. Second, based on the problem definition, the expert and the engineer work together on the knowledge acquisition process. Throughout this knowledge acquisition process, the KE and the expert interact several times. The purpose is for the KE to extract relevant expert knowledge from the HE to define the knowledge base (Fig. 5.1). Extracted knowledge must be concise, unambiguous, and targeted to the required purpose of the ES. It is important to remember that the ES will serve as a computer proxy of the expert. Therefore the more relevant knowledge is extracted, the more useful the ES will be.

Extracted knowledge, typically from the HE in the form of statements, is converted into facts and rules and integrated into the knowledge base. Communication and understanding between the expert and the engineer are vital for this task to succeed. The expert must be able to clearly communicate the knowledge (rules of thumb) used to solve problems in a specific domain, while the engineer, besides understanding such knowledge and the needs for the ES, must correctly “translate” such knowledge into a suitable representation

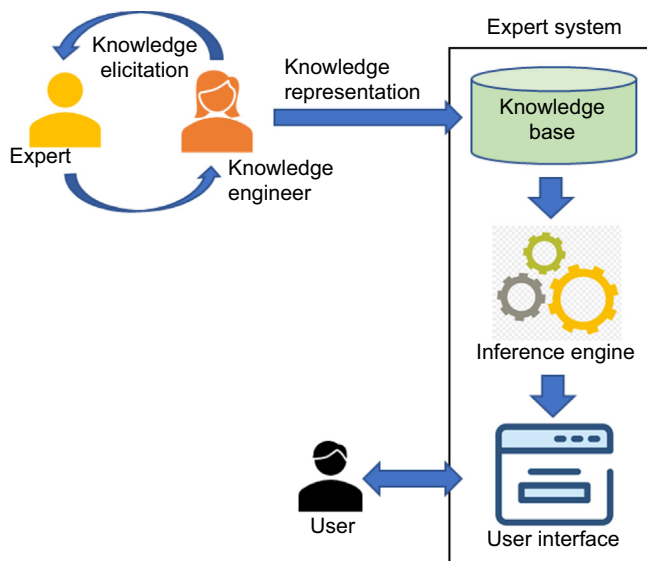


FIGURE 5.1 Schematic view of an expert system, and its interactions with expert, knowledge engineer, and user.

without any loss of heuristic knowledge. Oftentimes, the expert struggles to accurately and explicitly communicate all the knowledge required to perform specialized tasks. If we add a potential lack of understanding of the domain by the engineer, this may result in suboptimal representations of knowledge, reflected in inaccurate facts and rules that do not perform as expected.

In an ES the knowledge base is separate from the execution engine. The knowledge base contains facts and rules, and the execution engine applies the rules and reasons about the facts. Reasoning is a logical and heuristic process that takes into consideration notions such as actions to be taken under given circumstances, goals to be achieved, causality and dependencies among rules and facts. There is no room for improvisation or generalization; the engine simply processes all the rules that apply to the presented facts and produces an answer. As mentioned previously, if the gathered knowledge is ambiguous or incomplete, the results from the ES will potentially also be ambiguous and incomplete. Therefore it is essential that the knowledge base is as complete and unambiguous as possible so that the reasoning is sound.

Even though knowledge acquisition can be costly and time consuming, when done correctly, the benefit is improved decision quality in terms of fast, reliable, consistent, and efficient solutions. In addition, given the nature of the decision rules, ES provides answers about and explanations of the decision-making process. Getting an explanation about an answer is key in many domains, such as medicine, where, for example, getting a potential diagnosis for a patient could be as important as getting an explanation as to why that diagnosis is relevant.

There are two common approaches for representing knowledge in the knowledge base: frame-based systems and production rules. First proposed by Marvin Minsky in the 1970s, frame-based systems provide a natural structure to organize knowledge in terms of frames representing concepts or stereotypical situations.²⁸ The internal structure of a frame, that is, slots, stores all the necessary information about characteristics, attributes, usage, expectations, and so on. This means that all kinds of definitional and descriptive information can be included in a frame.

In terms of internal structure within a frame-based system, collections of frames are organized and, when relevant, interconnected to each other through relationships. Execution of the reasoning process on these systems matches a frame to a specific situation, using default values to fill unspecified aspects/slots of the frame.

Production rules are the most common KR method, particularly in medicine, MYCIN being the most famous example,⁴ and in business.^{26,27} In rule-based ESs, knowledge is represented by production rules. Each rule consists of an antecedent, or condition, and an action, or conclusion. The general form is as follows:

IF *condition* THEN *action(s)*

During the reasoning process, all the rules for which the conditions in their antecedents are satisfied are triggered and the respective actions executed. The inference engine orchestrates the rule execution order and resolves conflicts between rules. This process is mostly centralized and managed by rule properties to control the order in which rules are executed. This is achieved mostly through execution parameters and conditions, rule priorities, and execution flow.

When a user enters relevant facts about the case at hand, these facts are stored in a working memory. The inference engine applies one of two possible strategies, working

either *forward* (data-driven) or *backward* (goal-oriented) chaining rules from the knowledge base to the facts in the working memory. As rule conditions are satisfied and new facts are generated, these new facts are added to the working memory. This process continues until a goal state is reached and no more rules apply to the facts in working memory. These systems provide an explanation facility to explicate how the system triggered certain actions or arrived at a given recommendation. Explanations can be as simple as rule numbers or identifiers for triggered rules, or more elaborate, human-like explanations. Either way, this approach provides explicit information about the reasoning process.

In a *forward* (data-driven) chaining strategy, the inference process is driven by the facts stored in the working memory. The rule engine attempts to match the antecedent (condition) of each rule in the knowledge base using the available facts. Execution order is handled by the parameters explained previously, and, if needed, resolution strategies are applied. This type of reasoning strategy is normally used to solve open-ended problems such as planning.²⁹

Backward chaining strategy starts from an assumed goal or subgoal state—the THEN (conclusion) part of the rule. The execution engine tries to match the antecedent of the rule with the assumed conclusion. If such a match exists, its antecedent becomes the new subgoal, and the process repeats; otherwise the engine will attempt to prove other goal(s) until a goal state can be supported by the antecedent. In other words, this strategy works *backward*: it starts with a conclusion in a rule and tries to satisfy the conditions for that rule. If this does not work, it picks another rule conclusion and tries to satisfy its conditions with the available facts. The execution engine keeps trying until it finds such a rule, or no more rules are left to evaluate. This type of reasoning strategy is best used in diagnosis applications with a limited number of well-defined conclusions that can be checked.²⁹

5.3.2 Knowledge representation and management

The term “knowledge representation” encompasses all formalisms employed for appropriately represent, encode, and execute knowledge. The processes for eliciting, representing, and encoding knowledge normally are referred to as knowledge engineering. Ontologies and reference terminologies and vocabularies play a vital role in representing knowledge in a clear, succinct, unambiguous manner. During the modeling process, the KE must take into account all available evidence and information and transform it into unambiguous definitions. This is followed by the identification of relevant reference terminologies and vocabularies used to further describe the semantic meaning of the knowledge being encoded. This step provides further consistency to the representation efforts in that the meaning of the knowledge asset being represented is reinforced and enhanced by the meaning of the reference terminologies and vocabularies. Further, once knowledge assets are linked to other knowledge assets, relationships among existing and new concepts emerge, providing richer definitions that go beyond those initially ascribed to individual concepts. The outcome of this modeling process is either a human-readable, declarative representation or a computer-interpretable representation of the initial knowledge.³⁰ Computer-interpretable knowledge expressed as described previously can be easily incorporated into knowledge bases and reasoned upon by inference engines. Regardless of the

final representation, care must be taken to ensure that knowledge is concise, explicit, and unambiguous. This allows for an incremental approach to expanding knowledge while preserving consistency in the definitions and relationships among connected knowledge assets. An additional step to further enrich the semantics of encoded knowledge is to link knowledge assets to clinical domain ontologies. These ontologies provide additional mechanisms for inference and consistency checking and ongoing maintenance.³¹

Ontologies represent collections of formalized concepts that semantically augment knowledge assets. This is advantageous in terms of modeling, since ontologies not only allow for structuring concepts and establishing relationships, but when reason upon, they also provide a solid foundation for knowledge discovery.^{32–34} Further, structured, clear, and explicit knowledge endorsed by reference terminologies and vocabularies foster shareability and reusability of knowledge. Even when the representation of knowledge requires local customization (not ideal), having clear, semantically augmented definitions, facilitates the customization process and reduces potential misunderstandings.

Syntax is another key aspect of KR. The notation used to represent knowledge and describe concept hierarchies and maintenance mechanisms is the means for ensuring that a KR contains all the facts needed for the engine to traverse and reason upon the meaning of the knowledge. The syntax defines the elements, structure, and valid configurations of those elements in order to encode valid “sentences.”

There is a consensus on the need for standards for KR to address syntactic issues. However, there is a lack of general agreement on a common representation and the mechanisms for translating knowledge encoded in one formalism into another. Some of the approved standards for clinical KR that have had some traction over the years include Arden Syntax⁶ and GELLO.¹⁴

Arden Syntax is part of Health Level Seven International (HL7).³⁵ It was published by HL7 in 1999. Arden Syntax is considered a hybrid KR.^{6,36} It supports encoding of knowledge and definition of workflows. Arden Syntax is used to define medical logic modules (MLMs) comprising variable definitions and execution flow guided by decision logic. MLMs are self-contained and can be embedded and invoked at any point in workflows where a decision, represented in the MLM, is needed. The main limitation of Arden Syntax that so far has precluded the shareability of MLM is the “curly braces problem.” Within an MLM, calls to local data repositories are defined within curly braces. The notation for retrieving such data needed for the decision logic of the MLM is specific to the institution and system. This means that not only is the call to retrieve information tied to a specific syntax, but the data itself and its structure are also specific. This presents a major challenge when MLMs are shared, since this requires extensive recoding of the content within curly braces to the point that often it is better to define the MLM anew. Fig. 5.2 presents a fictitious example of an MLM for screening and notification of evidence of recent myocardial infarction. The MLM is divided into three main sections: maintenance, library, and knowledge. Note the data retrieval in an institution-specific format within the curly braces.

Another approved KR standard is GELLO: an object-oriented, platform-independent expression language for data querying, reasoning, and evaluation of logic expressions.¹⁴ In 2005 GELLO was approved as an international standard by ANSI and HL7.³⁸ GELLO is based on the Object Constraint Language (OCL) and is declarative in nature. OCL is an Object Management Group standard that can be used with any object-oriented data model.

```

Maintenance:
  Title:      Screen for positive troponin I;;
  Filename:   troponin;;
  Version:    1.40
  Institution: World famous medial center;;
  Author:     Robert A. Jenders MD, MS
  Specialist: ;;
  Date:       2013-04-30;;
  Validation: research;;

Library:
  purpose: screen for evidence of recent myocardial infarction;;
  explanation: triggered by storage of troponin result. sends
               message if result exceeds threshold;;
  keywords:   troponin; myocardial infarction;;
  citations:   ;;

Knowledge:
  type:       data-driven;;
  data:       troponin_storage:= event(storage of troponin);
               /* get test result */
               tp:= read last {select result from test_table where
                             test_code = 'TROPONIN-I';

               threshold := 1.5;;
               /* email for research log */
               email_dest := destination {'email', 'name' = abc@ucla.edu'};

               ;;
  Evoke:      troponin_storage;;
  Logic:      if (tp is not number) then conclude false;
               endif;

               if tp > threshold then conclude true;
               else conclude false;
               endif;

               ;;
  Action:     write "Patient may have suffered a myocardial infarction." ||
               "Troponin I = " || tp || " at " || time of troponin
               at email_dest;

               ;;
  Urgency:    50;;

end:

```

FIGURE 5.2 Sample Arden Syntax module (MLM) for screening for troponin. The MLM is divided into three sections: maintenance, library, and knowledge. Note the reference to institution-specific data source within the curly braces. *Source: Adapted from Jenders RA. Decision rules and expressions. In: Clinical decision support. Elsevier; 2014. p. 417–34.*³⁷

GELLO captures the specific features of clinical knowledge, for example, eligibility criteria required in guidelines and expresses it in a declarative, deterministic manner. GELLO uses the HL7 Reference Information Model (RIM) as data model. This allows the KE to build expressions that query data from a standard source in a standard representation and logically manipulate and reason upon them. GELLO also supports calculations and other formulae, thereby, facilitating close integration with other HL7 standards that use the RIM. Its generic nature allows it to be integrated into multiple applications. GELLO addresses the curly braces problem by facilitating the incorporation and use of standard vocabularies and

```

let lastTroponin: Observation = Observation → select (code=
  ("SNOMED-CT", "102683006")).sortedBy(effectiveTime.high).last()

let threshold : PhysicalQuantity = Factory. PhysicalQuantity("1.5", "ng/dl")

let myocardial infarction: boolean = if lastTroponin. value.greaterThan (thershold)
  then
    true
  else
    false
  endif

if myocardial infarction then
  send notification
else
  send alternative notification
endif

```

FIGURE 5.3 Same clinical knowledge for troponin as presented in Fig. 5.2. Note the succinctness and declarative nature of GELLO. Also note the reference to HL7 RIM (standard data model). GELLO expressions can be embedded into clinical workflows.

data model. Even though the declarative nature of GELLO is more difficult to grasp than the procedural structure of Arden Syntax, authoring tools can hide this complexity, allowing authors to create computable knowledge (Fig. 5.3).

5.3.3 Uncertainty, probabilistic reasoning, fuzzy logic

5.3.3.1 Uncertainty

In classic propositional logic, a statement is either true or false, but people often take the truth value as a matter of degree. As noted in a previous section, production rules are dichotomous, and, when applied, the antecedent (the statement embedded in the IF part of the rule) is either true or false, but nothing in between. The inference process has a pre-specified goal, and the process is *terminatable* whenever its goal is achieved. In the real world, this imposes limitations on certain application areas, because the nature of the data collected do not always fit a clear yes–no scenario, and therefore, dichotomous rules cannot be applied. There are facts we *believe* to be true or false. Hence, it is advantageous for a system to be able to reason under uncertainty and provide ways to represent this vagueness and ambiguity in a formal, well-defined language that can produce conclusions and justify them.

5.3.3.2 Probabilistic reasoning

Probabilistic reasoning arises from the need to quantify and reason under uncertainty. It is a richer, more expressive formalism that preserves the process of deductive reasoning from statements to reach a logically certain conclusion and incorporates probability theory to represent and handle uncertainty. It attempts to define a natural extension of traditional dichotomous logic, where the truth value of a proposition is extended from discrete, binary values to a [0,1] interval of continuous values, with binary logic as a special case. A conclusion is the consequence of the premises, but unlike propositional logic, the truth value is not dichotomous but a function of the conditional probabilities of the premises. Under this notion, only a conclusion under the special case will be absolutely true, while

any other potential conclusions will have a truth value based on the probabilities of the premises, and the one with the highest probability will be preferred.

Probabilistic reasoning interprets these probabilities as reasonable expectation,³⁹ or quantification of a personal belief.⁴⁰ The following are definitions needed in order to understand probability theory.

Boolean connectives (1) negation, (2) conjunction, and (3) disjunction are extended into probability functions to handle the probabilities of the premises being evaluated. Boolean connectives become probability functions of events A , B as follows:

1. Negation: $P(\neg A) = 1 - P(A)$, where $P(\neg A)$ is the probability of A not occurring.
2. Conjunction: $P(A \cap B) = P(A) \times P(B)$, where A and B are independent of each other.
3. Disjunction: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, where $P(A \cup B)$ denotes the probability of either A or B happening, but not both.

When a proposition A can only have two truth values: true or false, *a priori* or *unconditional probability* $P(A)$ represents the probability of that proposition A being true.⁴¹

The sum of all the probabilities x_i of a *random variable* X is always 1 and is denoted by:

$$\sum_{x \in \{x_1, \dots, x_n\}} P(X) = 1 \quad (5.1)$$

The *conditional probability* of A given the prior occurrence of B , denoted by $P(A/B)$, is defined as the ratio of the joint occurrence (conjunction) of A and B to the probability of B :

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} \quad (5.2)$$

Similarly, the conditional probability of B given A is denoted by:

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} \quad (5.3)$$

Manipulating Eqs. (5.2) and (5.3), respectively, we obtain:

$$P(A \cap B) = P\left(\frac{A}{B}\right)P(B) \quad (5.4)$$

$$P(A \cap B) = P\left(\frac{B}{A}\right)P(A) \quad (5.5)$$

By combining Eqs. (5.4) and (5.5), we derive:

$$P(B|A)P(A) = P(A|B)P(B) \quad (5.6)$$

and

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)} \quad (5.7)$$

which is Bayes' theorem on conditional probability. For a more in-depth description of uncertain knowledge and reasoning, see Ref. [41].

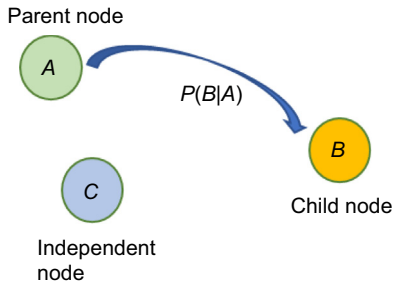


FIGURE 5.4 Conditional probability $P(B|A)$ in a directed arc connected from A to B . A and B are parent and child nodes, respectively. Node C is not connected to either A or B and therefore is conditionally independent.

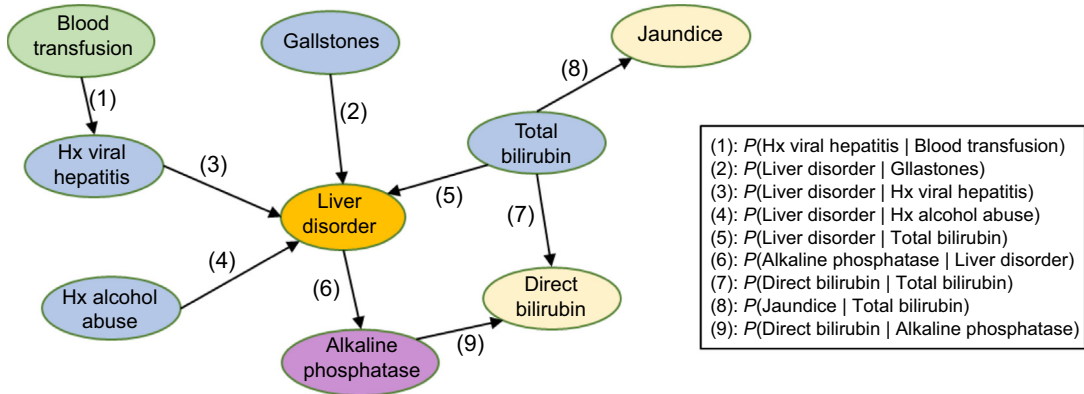


FIGURE 5.5 Simplified Bayesian network for diagnosis of liver disorder. Source: Adapted from Onisko A, Druzdzal MJ, Wasyluk H. A Bayesian network model for diagnosis of liver disorders. In: Proceedings of the eleventh conference on biocybernetics and biomedical engineering. CiteseerX; 1999.

Bayesian network is a probabilistic, directed acyclic graphical (DAG) model that represents a set of variables and their conditional dependencies, where the strength of these dependencies is denoted by the conditional probabilities. Nodes in a graph represent variables in the Bayesian sense, and arcs depict the conditional dependencies (joint probability distribution) between the parent (A) and child (B) nodes. Conditionally independent nodes (C) are not connected to other nodes (see Fig. 5.4). Formally, the structure of a DAG represents a factorization of the joint probability distributions of the considered variables.⁴²

Bayesian networks are well suited to represent the probabilistic relationships between diseases and symptoms. Given that a Bayesian network is a complete model of a set of variables and their relationships, it can be used to compute the posterior distribution of variables based on available evidence. It applies Bayes' theorem to perform probabilistic inferences. For example, a Bayesian network could be used to compute the probabilities of the presence of a disease based on the presence of certain symptoms. As shown in Fig. 5.5 (simplified from Ref. [43]), multiple factors are taken into consideration for a potential diagnosis of liver disorder. Based on available evidence, for example, symptoms and findings, objective evidence observed by physicians, and laboratory test results, the applicable conditional probabilities are calculated to obtain a potential diagnosis.

Compared to other probabilistic reasoning approaches, Bayesian networks are more efficient, though computational costs can easily escalate for complex models. In addition to the computational expense, unknown probability values and inconsistent probability assignments can hinder the applicability of this methodology.

5.3.3.3 Fuzzy logic

Fuzzy logic is an extension of dichotomization of one of the fundamental underlying concepts of classic propositional logic proposed by Zadeh.⁴⁴ In propositional logic, truth values are binary, and we should be able to unequivocally determine, based on the characteristics of an object, whether it belongs to a set or not; that is, an object x is either a member of a set A or not (Eq. 5.8).

$$A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases} \quad (5.8)$$

However powerful this concept may be, the condition defining the boundaries of a set is very rigid. Fuzzy sets extend the binary membership of a conventional set into a continuous interval $[0,1]$, where the membership values express the degree of compatibility of an object with the distinctive properties or characteristics of the collection it belongs to. The membership value can range from 0 (complete exclusion) to 1 (complete membership). Thus a fuzzy set A is a set of ordered pairs defined by⁴⁵:

$$A = \{ (x, \mu_A(x)) : x \in X \} \quad (5.9)$$

where x is any object of a universal set X and $\mu_A(x)$ is the degree of membership of the element x in A . The following example of the resting heart rate (RHR) of adult patients at risk of developing complications illustrates these concepts.

Let X be the universal set of RHR measurements for adults where a RHR x in X could have any value in the range of $[0,250]$ beats per minute, with three fuzzy sets to classify patients, for example, low, normal, and high. As part of this fictitious system, there are alerts to warn when a patient's heart rate deviates from normal. All possible heart rates in the interval $[0,250]$ are members of each fuzzy set, though with different membership values in the interval $[0,1]$. For example, an RHR = 70 is a member of all three sets, but the membership functions $\mu_{\text{LOW}}(70)$, $\mu_{\text{NORMAL}}(70)$, $\mu_{\text{HIGH}}(70)$ will have very different membership values for the same heart rate (Fig. 5.6).

In fuzzy systems, membership functions can be continuous, and expand over a continuous spectrum of values (Fig. 5.6), or discrete, sampling values at regular intervals over the whole range of possible values (Fig. 5.7).

Although any continuous function of the form $A:X \rightarrow [0,1]$ can be used as membership function of a fuzzy set, the most commonly used are: triangular, trapezoidal, S-membership, and Gaussian functions.

A triangular membership or bell-shaped function determines the degree of membership by comparing a given value x against upper and lower values a, b and a modal value m as shown in Eq. (5.10). The graphic representation of this type of function is depicted in Fig. 5.8A.

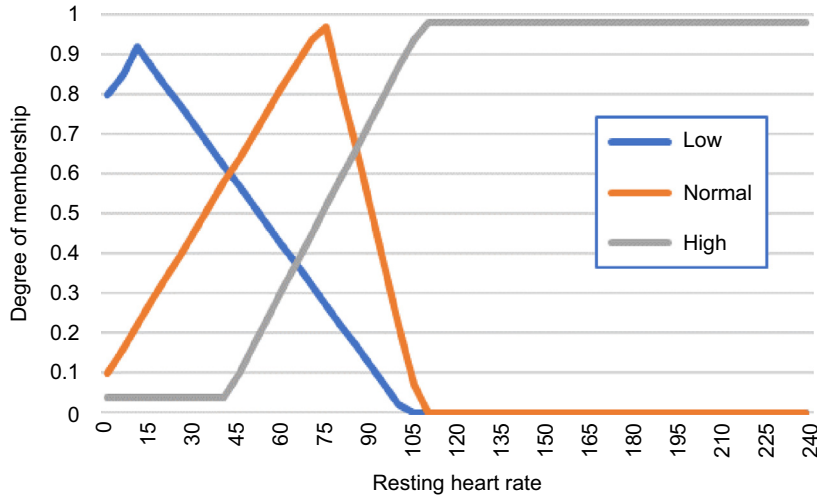


FIGURE 5.6 Membership curves for the fuzzy sets low, normal, and high. The x -axis denotes the resting heart rate for adults. The y -axis denotes the degree of membership of the given fuzzy sets at different heart rates.

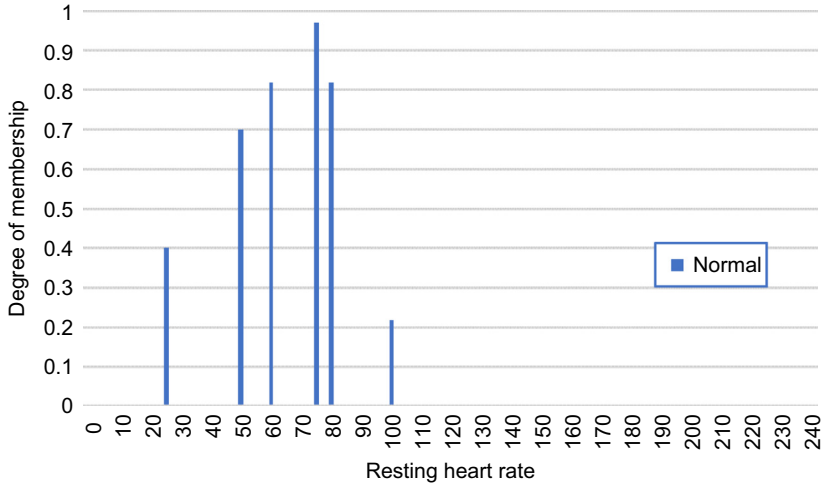


FIGURE 5.7 Nonuniform sampling of discrete resting heart rate values at nonuniform intervals for the normal fuzzy set.

$$A(x) = \begin{cases} 0 & \text{if } x \leq a \\ \frac{x-a}{m-a} & \text{if } x \in [a, m] \\ \frac{b-x}{b-m} & \text{if } x \in [m, b] \\ 0 & \text{if } x \geq b \end{cases} \quad (5.10)$$

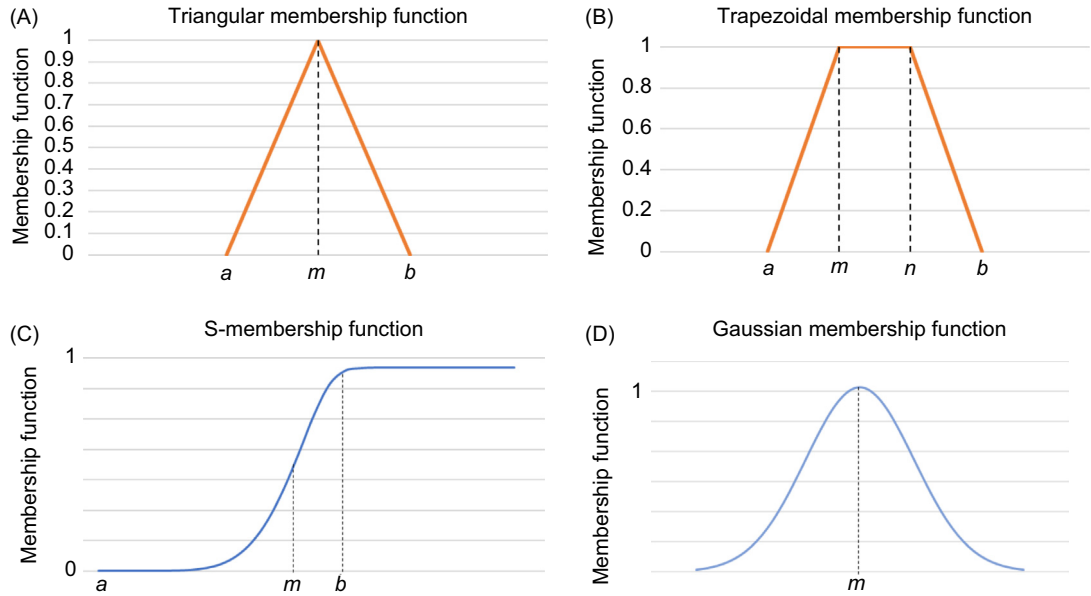


FIGURE 5.8 Typical membership functions: (A) triangular; (B) trapezoidal; (C) S-membership; and (D) Gaussian.

A trapezoidal membership function is typically used to represent the fuzzy linguistic conditions *neither so high nor so low*.⁴⁵ In the heart rate example, this function could be used to express that a given heart rate is *neither too high nor too low* for those values between the threshold boundaries determined by m, n , calculated by Eq. (5.11). See Fig. 5.8B.

$$A(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{m-a} & \text{if } x \in [a, m] \\ 1 & \text{if } x \in [m, n] \\ \frac{b-x}{b-n} & \text{if } x \in [n, b] \\ 0 & \text{if } x > b \end{cases} \quad (5.11)$$

An S-membership function has a smoother slope. In the previous example, it could be used to calculate the membership for the fuzzy set *high*. One typical form of the S-function is depicted in Fig. 5.8C, and the curve is calculated by Eq. (5.12):

$$A(x) = \begin{cases} 0 & \text{if } x \leq a \\ 2 \left(\frac{x-a}{b-a} \right)^2 & \text{if } x \in [a, m] \\ 1 - 2 \left(\frac{x-b}{b-a} \right)^2 & \text{if } x \in [m, b] \\ 1 & \text{if } x > b \end{cases} \quad (5.12)$$

A Gaussian membership function has a wide application on fuzzy sets. In our example, this function could be used to calculate the membership for the fuzzy set NORMAL, with mean value of, for example, 60. To shape the curve, we can experiment with different values for the variance; the smaller the value, the higher and sharper the curve around the mean value. A typical Gaussian curve is depicted in Fig. 5.8D, and it is calculated by Eq. (5.13).

$$A(x) = e^{-k(x-m)^2} \quad \text{where } k > 0 \quad (5.13)$$

5.4 Applications

Studies have shown that appropriately implemented medical ESs can enhance health-related decisions and actions with the right knowledge presented at the right time, and therefore reduce medical errors and improve quality of care. A wide range of tools and interventions have been developed and implemented over the past 50 years, either as stand-alone or as part of integrated solutions. Key CDS interventions include diagnostic support, therapy recommendations, order sets, medication alerts, and preventive care interventions and reminders. Some institutions have also used ESs for billing purposes, for example, by recommending a plan and treatment options based on the health-care needs of the patient and financial needs of the institution. In the following sections, we show some examples of medical ESs.

5.4.1 Computer-assisted diagnosis

An accurate diagnosis is important because it helps a physician choose an appropriate therapy for a patient and may also reduce diagnostic errors. Medical diagnosis is often challenging and involves complicated reasoning processes, as it requires collection of various pieces of information (e.g., signs, symptoms, medical history, and lab test results) for differential diagnosis; however, the collected information may be nonspecific, causing uncertainty. Ledley and Lusted² proposed that certain mathematical techniques, including symbolic logic, probability, and value theory, can help our understanding of the reasoning behind medical diagnosis and the choice of an optimum treatment. Computers can help clinicians collect and process relevant information during the diagnostic process to search through a collection of possible diseases to make a more precise diagnosis and a more scientific decision about the treatment plan. In the 1970s the Leeds Abdominal Pain System was built based on Bayesian probability theory and used sensitivity, specificity, and disease-prevalence data for various signs, symptoms, and lab test results to calculate the probability of the diagnosis of abdominal pain.⁴⁶

MYCIN, one of the most influential early ESs, uses a rule-based approach and symbolic reasoning techniques rather than numerical calculations. MYCIN was developed as a thesis project by Edward Shortliffe at Stanford University that aimed to help physicians select an antimicrobial therapy for patients with infections.^{4,19} Clinical knowledge of bacterial infections is represented as a set of production rules, where each rule has a certainty factor

associated with it. MYCIN is goal driven; it uses backward chaining reasoning from a goal or conclusion to conditions that establish it. It assesses patient-specific data entered by the physician to generate conclusions and therapeutic advice and also explains its reasoning and decisions when asked to do so. MYCIN was designed to keep its knowledge base separate from the reasoning engine, so the domain-independent portions of MYCIN, also known as EMYCIN, or “Essential MYCIN,” were used to develop other ESs. For example, PUFF was a system that could process lung function data, recommend four possible diagnoses, and generate pulmonary function reports.⁴⁷ The agreement between the PUFF system and two physiologists was 92%. ESs built to provide consultation in diagnosing difficult and rare pulmonary diseases can be found in this review.⁴⁸

Other ESs built to provide consultation in medical diagnoses across different clinical settings and domains include HELP (*health evaluation through logical processing*),⁴⁹ INTERNIST-I,⁵⁰ Quick Medical Reference,⁵¹ CADUCEUS,⁵² and DXplain.⁵³ DXplain, a system developed at the Laboratory of Computer Science at the Massachusetts General Hospital, generates ranked differential diagnoses based on user input of a patient’s symptoms, laboratory results, and other clinical findings using a modified form of Bayesian logic to derive clinical interpretations.⁵³ It has been used mainly for medical training.

5.4.2 Computer-assisted therapy

Some ESs were built to provide CDS for medical therapy. For example, MYCIN recommended antibiotics for bacterial infections.¹⁹ ONCOCIN,⁵⁴ an oncology protocol management system, was designed to assist physicians in chemotherapy for cancer patients. The protocols for cancer treatment were represented and coded in ONCOCIN’s knowledge base. Physicians interacted with the “interviewer” to review and enter the patient’s data and received therapy and test recommendations generated by the “reasoner.” An antibiotic consultant program was integrated with a hospital information system, HELP system, at LDS hospital in Salt Lake City.⁵⁵ The program accessed the patient’s medical records and used patient- and infection-specific information provided by physicians to determine the patient’s “most likely” pathogens. The likelihood of the pathogens was estimated using data from the previous 5 years and the most recent 6 months from patients with similar characteristics. The computer displayed the five antibiotic regimens most likely to be effective for all the pathogens and suggested an appropriate antibiotic regimen (e.g., based on the infection site). The program also checked patients’ allergy history and renal function using logic rules. The evaluation of the program showed a 17% greater pathogen susceptibility to an antibiotic drug suggested by the computer program versus physicians.

Studies have also focused on dosing of medications for patients with certain clinical conditions to optimize prescribing behavior and reduce medical errors. A group of researchers at the Brigham and Women’s Hospital built an algorithm within a computerized physician order entry system for adjusting drug dose and frequency in patients with renal insufficiency.⁵⁶ The researchers created a knowledge base containing a list of medications that were renally cleared and/or nephrotoxic and the optimal adjusted dose/frequency of each medication for different levels of renal insufficiency. Compared to the control, the intervention resulted in the selection of 15% fewer inappropriate doses and 24% fewer inappropriate

dose frequencies. Other drug-dosing systems assisted clinicians in maintaining therapeutic theophylline level^{57–59} or warfarin dosing during initiation and follow-up maintenance therapy.^{60–63}

A computer-assisted antibiotic-dose monitor was designed at the LDS hospital in Salt Lake City, Utah, in 1995, which was used as a daily surveillance tool to monitor for appropriate antiinfective dosages.⁶⁴ The program used computer-based patient records to screen patients in the hospital who were receiving any of the studied antibiotics. Then, the program calculated an optimal 24-hour dosage for the prescribed antibiotic, adjusted by the patient's renal function test results and based on clinical guidelines. The program printed out the list of patients who were determined to possibly be receiving excessive dosage for the clinical pharmacists to review. The pharmacist then contacted patients' physicians, when needed, to discuss the possible need for a change in the dosage.

In medicine, physicians often fail to order tests or treatments (so-called corollary orders) needed to monitor or ameliorate the effects of other tests or treatments. For example, ordering Coumadin should trigger a decision to monitor prothrombin time and international normalized ratio. A group of researchers at Regenstrief Institute built computer-generated reminders to suggest corollary orders to physicians while they were writing their orders to reduce errors of omission.⁶⁵ They conducted a randomized, controlled clinical trial and found that intervention physicians ordered the suggested corollary orders in 46.3% of instances when they received a reminder, compared with 21.9% compliance by control physicians. This study showed that such reminder interventions can be an efficient means for reducing reliance on memory and improving adherence to practice guidelines.

5.4.3 Medication alert systems

Adverse drug events (ADEs) account for 19% injuries in hospitalized patients. Medication alert systems focus on identifying potential medication-related adverse events in various levels of severity.⁶⁶ These alerts are easily incorporated into clinical workflows and produce timely notifications to help clinicians prescribe the most appropriate medications based on the clinical status of a patient. Alerts focus on drug–drug interactions, medical condition–drug interactions, drug–allergy checking, and dosage adjustments. They monitor clinical events indicative of possible occurrences on ADEs by checking on patient medical conditions and notifying physicians about them so that physicians can take additional steps to address issues. For example, a patient with renal failure may require a dosage adjustment on a medication about to be ordered that is metabolized by the kidneys. These systems also monitor for potential drug–drug interactions by checking the current medication list of a patient and warning the ordering physician about potential antagonistic events when a new medication to be ordered adversely interacts with a current medication (e.g., warfarin and aspirin) and drug–food interactions by advising the physician to notify patients of foods that must be avoided while taking a medication (e.g., statins and grapefruit). They also monitor the impact of medication orders by checking on changes in laboratory tests associated with those medications, for example, drug-induced immune hemolytic anemia or age-related liver toxicity of acetaminophen.

Bates et al. evaluated the efficacy of computerized physician order entry and a team intervention targeted the administration and dispensing of drug on prevention of serious medication errors.⁶⁷ The computerized physician order entry system provides a menu of medications from a formulary and default doses and a range of potential doses for each medication. It also suggests relevant laboratory tests at the time of ordering and conducts some drug-allergy and drug-laboratory checking. The researchers reported 55% decrease in nonintercepted serious medication errors and 17% decrease in preventable ADEs.

5.4.4 Reminder systems

Patient-specific reminder systems are mostly divided into four categories: health maintenance, expensive medication reminders, chronic disease care/management, and therapeutic recommendations. Reminder systems identify patients for whom upcoming (almost due) and overdue maintenance screenings and laboratory procedures need to be scheduled. They also include notifications relating to alternative therapeutic interventions and cheaper alternative medications available in the formulary. These systems generate notifications addressed to primary care physicians and health-care providers alerting them about upcoming due dates for preventive and monitoring actions and procedures that need to be scheduled, for example, a woman due for Pap smear, or a regular lab test for a diabetic patient monitored for hemoglobin A1c.

Triggered reminder rules generate notifications that are either printed on a patient encounter summary, informing both the physician and the patient of upcoming preventive actions, or presented to the physician as part of the CDS embedded in careflows. Following an on-screen notification, physicians or other assigned health-care staff follow up on the recommendation. The system generates letters and notifications to patients informing them of the need to schedule upcoming or overdue preventive actions.

5.5 Challenges

Although great progress has been made, many challenges still exist around the life cycle of ESs, from knowledge acquisition and management (including the process of designing, testing, deploying, and maintaining rules), to integration with EHR and workflow, knowledge sharing, and performance evaluation. Clinician acceptance, financial obstacles, and impact on institutional culture are also important factors to be considered. The evidence of the effectiveness of ESs in improving CDS and the quality and efficiency of care has been mixed. A 2005 systematic review reported that CDS systems improved provider performance in 64% of the studies and patient outcomes in 13% of the studies.⁹ Some reviews were less optimistic about the effects of CDS, calling for more evidence to demonstrate the cost-effectiveness of these systems.⁶⁸

5.5.1 Workflow integration

Workflow integration refers to processes, content, or frameworks set in place to support workflow itself. Workflow is defined as a series of tasks where jobs and processes can be

streamlined by the seamless incorporation of tools to aid and enhance the outcome of tasks, not as punctual, isolated elements, but as components of a whole. It is what connects the dots and how best to optimize those steps in order to improve job performance.

Collaborative tools support creation, management, and deployment of knowledge. They provide an excellent platform for consistent, trackable, reproducible knowledge-related activities aligned with best practices and recommendations. In terms of knowledge, collaborative tools allow for modularity and specialization. Knowledge assets can be customized to address specific needs; they can be self-contained, thereby facilitating integration.

There are a wide variety of workflows in clinical systems. In many cases, these workflows have knowledge assets in common that could be centrally managed in a collaboration platform and deployed into those systems as required. Alternatively, execution of those assets may be triggered at a specific point during a workflow as adjuncts to assessment, diagnosis, treatment, or prevention tasks. Those assets could be plugged in at critical points of a workflow, and, based on specific conditions, trigger actions relevant to the task at hand.

As seen in the previous section, computer applications provide valuable decision support at the point of care in the form of medication, therapy, treatment, dosage recommendations, and interventions. Integration of these knowledge assets requires them to be self-contained, so, for example, a rule advising about a medication dosage, or a medication order set containing relevant medication to treat a disease can be plugged into a medication ordering workflow.

In order to maximize the benefits of workflow integration activities, it is necessary to have well-defined, consistent, and reproducible processes for knowledge creation and maintenance, as well as knowledge deployment mechanisms. Having all these processes in place fosters implementation and applicability of knowledge-based interventions, particularly those requiring continuity during and after care setting transitions. Further, when workflow is integrated across applications, the benefits of each application improve. Knowledge and data are seamlessly shared between systems and applications, eliminating the need for separate, disparate applications to complete a process.

5.5.2 Clinician acceptance and alert fatigue

When one is exposed to a high volume of warnings, reminders, and recommendations, especially those who do not require escalation, one may become desensitized to these CDS interventions. In medicine, although some expert and CDS systems have been shown to improve the safety and quality of patient care, a high number of alerts and reminders presented to clinicians on a daily basis can result in them experiencing alert fatigue. Clinicians may pay less attention to or ignore both important and unimportant alerts warnings. Ignoring clinical alerts can lead to patient harm and other unintended consequences. It is reported that clinicians override 49%–96% of medication alerts and that many overrides are justifiable. The signal-to-noise ratio reveals significant problems; providers see and address more than 100 alerts to find one that could prevent one ADE.⁶⁹ One study collected 10 years of drug allergy alert data from two large academic hospitals in Boston and found that the overall override rate was 84%.⁷⁰ The most common override

reason for drug allergy alerts (~50% of the cases) was that the patient had previously tolerated the medication. In addition, clinicians were more likely to override repeated alerts that appeared two or more times than first time alerts (89.7% vs 77.4%, respectively). While most overrides may simply indicate low-value CDS recommendations that do not provide useful decision support, many alerts/recommendations are overridden in unsafe situations, such as in patients with histories of severe adverse drug reactions or contraindicated clinical conditions.

Because not all alerts need to be interruptive, to reduce overalerting, Paterno et al. designed and implemented a three-tiered presentation of drug–drug interaction alerts based on the level of severity.⁷¹ Level 1 alerts were the most serious, considered life threatening and required the clinician to either cancel the current order or discontinue the preexisting order. Level 2 alerts were less serious but still required action by the clinician (e.g., canceling the drug or providing an override reason). Level 3 alerts were the least serious and comprised the largest proportion of alerts. Level 3 alerts were presented as informational only and required no clinician action. Their study showed that, at the tiered site, 100% of the most severe alerts were accepted, compared to 34% at the nontiered site; moderately severe alerts were also more likely to be accepted at the tiered site (29% vs 10%). While a tiering approach has shown promising results in reducing overalerting generally, it has not been applied to other CDS domains.

Studies have shown that to increase clinical acceptance, it is important that a system can provide actionable recommendations rather just assessments. A system should explain its reasoning and/or provide evidence and encourage the user to record a reason when not following the advice generated by the system. When feasible and appropriate, it is also important for institutions to implement CDS surveillance systems to monitor inappropriate or abnormal alerts and recommendations. Web-based, graphic dashboards have been shown to be useful for monitoring CDS volume and trends over time. Poorly performing CDS alerts and recommendations should be identified for improvement. Other areas for improvement include EHR documentation and data quality, alerting mechanism (e.g., by considering contextual information, such as whether the alert was overridden in the past by the clinician), and hospital policies and guidelines.⁷²

5.5.3 Knowledge maintenance

Despite the ever increasing demand for computer-interpretable knowledge,^{73,74} the creation, testing, and maintenance of knowledge remains a manual, error-prone, and resource-intensive process. It is well known that creation of knowledge requires a detailed review of potential content and its translation into executable knowledge that could be integrated into clinical systems. The nature of clinical knowledge is complex, and it normally requires definitions that involve associations to other knowledge elements and reference terminologies. This creates multiple interdependencies that are difficult to track and manage. For example, a medication-related rule can easily become inoperable if the medication is removed from a formulary or is updated with a new medication; similarly, a rule advising about a diagnostic test will no longer trigger if the diagnostic test is superseded by a newer one, or if its reference value is updated. This problem becomes even more apparent when knowledge assets

are part of collections maintained by third-party users. In this case, reusability and maintenance require a degree of customization, thereby creating a disconnect between the source and the deployed versions. Extensive customization to create complex “localizations” required to incorporate contextual constraints to improve specificity of an intervention¹³ or adapt to workflow and staffing requirements are problematic and difficult to handle. The eagerness of institutions to implement and deploy new knowledge rapidly disappears as they realize the long-term maintenance implications. Therefore prior to implementation and deployment, it is important to set well-defined processes for creation, documentation, and maintenance of knowledge, so that subsequent changes are trackable and validated with evidence, decisions are justified and explained, and the whole process is transparent. Well-documented processes and decisions are more efficient, wasting no valuable time and resources and supporting the evolving nature of clinical knowledge.³⁰

5.5.4 Standard, transferability, and interoperability

Health-care knowledge is particularly complex and driven by extensive reliance on constantly evolving decision practices. This complexity arises not only from the intricate nature of medicine, but also from fragmented processes and incompatibilities in KRs and exchange mechanisms. From a knowledge engineering perspective, many of these problems can be mitigated with formal, well-defined KR methodologies and tools. Internally, organizations use several approaches to standardize knowledge repositories, practices, and procedures. However, wide exchange across teams and institutions can only be achieved through concerted collaboration and adoption of common tools and standards. Even though standards are, and have been, used for KR, the actual structure of the knowledge and descriptive attributes considered important widely vary from place to place. As a result, there is no guarantee that knowledge engineered at one institution, even if aligned with standards, would be transferable without loss of consistency into another institution. Without consistency, knowledge transferability and interoperability have diminished dissemination.

Over the past few years, standardization efforts have been intensified in terms of interoperability and exchange. The emphasis has shifted from creating standards for KR to standards that enable knowledge sharing and exchange.¹⁷ Government initiatives such as Meaningful Use have driven efforts toward developing standards for sharable knowledge. As a result, health-care institutions have been moving toward adopting and implementing systems that are more compliant with interoperability standards.

Despite these efforts, challenges remain. However, it is important to acknowledge that existing standards can be used as a lingua franca for knowledge exchange. In fact, each institution could maintain their internal representations, if their knowledge assets align with standard reference terminology vocabularies to preserve the semantics. Driven by semantics, mismatched knowledge can be reconciled into a common, agreed upon interchange of standard representation. Knowledge can be leveraged by its meaning, and the syntax adjusted accordingly, allowing for sustainable translations to and from interchange standards.

Alternatives for information exchange have surfaced over the past few years. The HL7 Clinical Document Architecture—release 2 was selected by the Office of the National

Coordinator as one of the data exchange standards for Meaningful Use.⁷⁵ However, exchanging whole complex documents turned out not to be an ideal approach. As a result, HL7 Fast Healthcare Interoperability Resources (FHIR)⁷⁶ emerged as a simpler approach that exposes discrete health-care data elements (e.g., patients, admissions, medication, and laboratory test results) as services that can be handled through URLs. FHIR provides the mechanisms for exchanging data between health-care applications. It facilitates interoperability between legacy and disparate systems, eases delivery of health information to providers, and allows integration of third-party applications into existing systems. The biggest challenge of FHIR is that it is possible for two different applications to implement different versions of FHIR with structurally different data elements and services. This means that the systems are not interoperable—they cannot exchange information because the representations are not compatible and cannot be consumed by the other system. Further, given the flexibility of representation of FHIR, it is possible to implement partial application programming interfaces, resulting in mismatched, partial, and incompatible representations. FHIR falls short of the interoperability expectations due to the lack of rigor in its representation mechanisms. To mitigate its limitations, an ontology-based analysis with formal mapping for correctness is currently under development.

5.6 Future directions

There are multiple opportunities for ES applications in clinical domains. However, important barriers mentioned in this chapter need to be overcome before wider, more efficient, and reliable applicability can occur. Broader incorporation of sophisticated, well-designed decision rules into clinical systems and workflows requires moving away from highly customized representations that only serve a narrow niche into high-quality interoperable assets that can be assembled into more complex, yet manageable structures that could be deployed into multiple systems and institutions. An integrated approach for implementation, deployment, and execution of knowledge assets must include sound infrastructure and policies for knowledge design, creation, and maintenance that persist throughout the life cycle of the knowledge assets.

There are three key technical aspects for the development of shared knowledge. The first is a common data schema and information model that serves as *lingua franca* when institutions maintain their own data structures and internal representations. The second is reference terminologies, vocabularies, and ontologies to semantically augment knowledge assets and data elements. By semantically augmenting assets, sharing knowledge and data elements goes beyond sharing syntactically well-formed assets; interoperability and exchange becomes a process of sharing meaning, which not only validates the syntactic definition of assets but also provides augmented meaning to ensure that mappings and “translations” are accurate to the source. The third is a common decision logic language and decision-making frameworks that provide the standard means for a consistent representation of executable knowledge. Knowledge should be integrated into workflows but should not be embedded in procedural code. Proper invocation mechanisms should exist to insert decision knowledge at critical points of workflow execution, but at the same time, remain independent. Clinical knowledge should be presented in a standardized format,

both human and machine interpretable, so system developers can produce the information in a way that front-end users can readily understand, assess, and apply it. Thus, a combination of a common data model, semantically augmented knowledge, and a common decision logic language are three key elements necessary if wide dissemination of consistent, reliable decision knowledge is to be achieved. Current efforts must align to these recommendations if they are to succeed in terms of true interoperability and shareability.

Understanding patterns found in data can leverage the rapid development and progress of data sciences and AI technologies. We need to understand the input-to-output process of data science and AI classification algorithms. However, this is a nontrivial process that requires identifying and elucidating valid, novel, and potentially useful patterns in data. In addition, we need innovative methods to improve reasoning mechanisms under challenging circumstances, as it is the case in medicine. If we aim at a wider applicability of these techniques, it is vital that such algorithms are capable of explaining how and why they provide an output (e.g., recommendation) based on some input data. Modern medical practice has emphasized EBM from well-designed research and the use of clinical guidelines to optimize clinical decision-making. However, EBM is hypothesis driven which may be influenced by various biases and constraints. In addition, there is a lag between when the research is conducted and when the evidence is properly applied. Therefore combining explanations from understood patterns in data and EBM may leverage both hypothesis- and data-driven methods into a more holistic approach that benefits from both approaches.

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